Sentiment-Driven Trading Strategy Design:
A Backtesting Study of the Hang Seng Index Using Social Media Signals

Fung Lok Lam
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Section 1: Introduction

Commissioned by a Hong Kong-based fintech firm, this report aims to design, evaluate and subsequently recommend a sentiment-based Hang Seng Index (HSI) trading strategy. Sentiment data is provided by the firm, a proprietary dataset containing daily pre-market social media poll which user predict whether the HSI would rise or fall the next day. The dataset spans over three years from 24 February 2022 to 12 March 2025. User responses are only gathered on trading days.

Figure 1 illustrates the daily pre-market Up Vote %, representing the percentage of users that speculate a rise in the HSI. To visualize overall sentiment dynamics in different time frames (2, 5 and 10 days), 2-day, 5-day, and 10-day moving averages (MA) are plotted alongside raw sentiment values. The sentiment and its MA lines are plotted shown in relation to the HSI's daily candlestick chart for the same period.

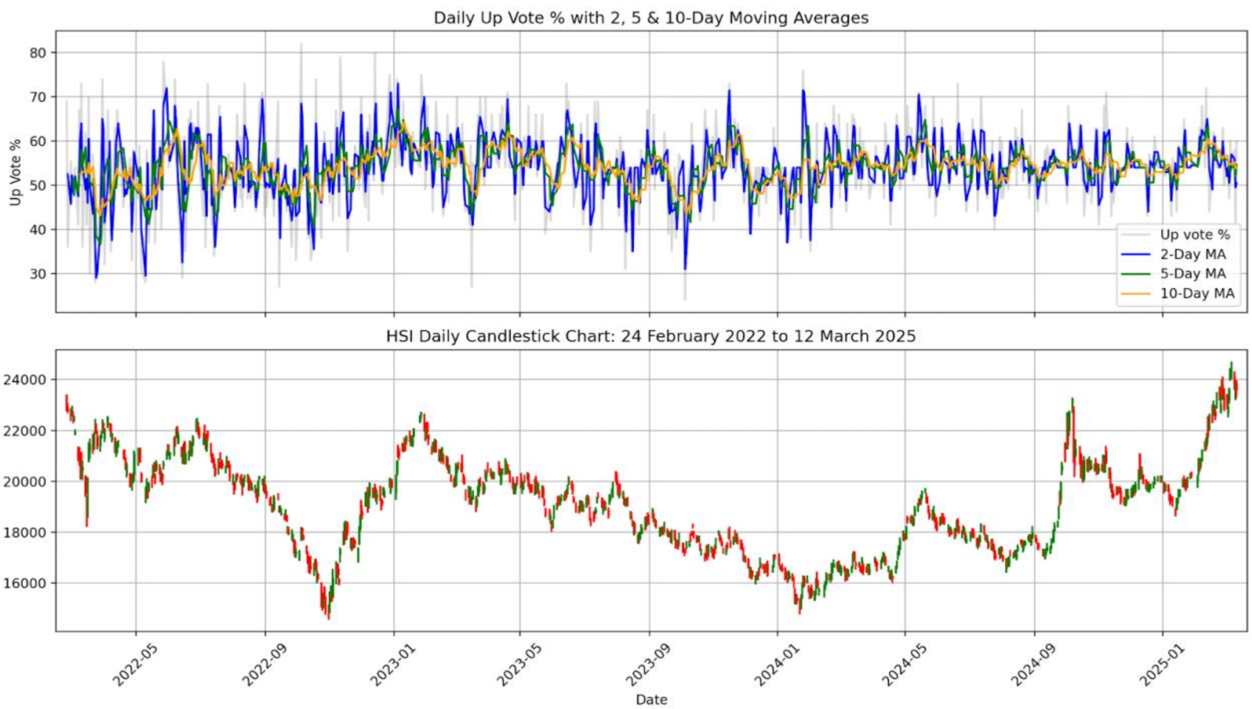


Figure 1: Up Vote % and HSI Trends

There is a rich collection of literature and practices in the market that theorize and use sentiment data for market trend predictions. The breadth of such literature and practices leads to a confusing, and contradictory conclusions on the effectiveness of different, and sometimes the same trading strategy. While some studies conclude that crowd behavior reflects forward-looking insights or momentum, others warn about risks of noise, herd behavior, or overreaction.

This report adds to the existing body of literature on behavioral finance and sentiment-based trading strategies. Three strategies based on sentiment-based logic are developed and tested via backtesting against the actual HSI performance. The aim is to: (1) identify if sentiment, particularly when pre-filtered through specific price-action rules can serve as reliable signals for daily trading decisions, (2) how these strategies perform relative to passive investment benchmarks and (3) recommend a new HSI trading strategy to the firm.

Section 2: Trading Strategies and Rationale

Three trading strategies were developed, with an emphasis on sentiment signals referencing existing trading theories. See Figure 2 below for details on each strategy's logic, theoretical context, and rationale.

Strategy	Logic	Theoretical Context and Rationale
Basic Buy in Positive Sentiment	**Buy when:** Today's pre-market Up Vote (%) reaches 55% or higher. **Sell when:** Today's pre-market Up Vote (%) reaches 50% or lower.	A basic strategy assuming a simple relationship between sentiment and HSI price. High pre-market Up Votes is an indicator of optimistic outlook by potential investors. Theoretically this increases demand for HSI shares, leading to increased share prices. Likewise, a low pre-market Up Vote % indicates pessimism or panic, leading to selloffs, hence slumped share prices.
Contrarian Buy in Extreme Fear, Sell in Euphoria/Extreme Greed	**Buy when:** Today's pre-market Up Vote (%) reaches 35% or lower. **Sell when:** Today's pre-market Up Vote (%) reaches 65% or higher *****. ***** Except in bull market regimes (when HSI closing price > HSI MA30).	This strategy is directly based on principles in *contrarian trading*, famously advocated by Warren Buffet and Charles T. Munger. It assumes that market overreactions often precede reversals. A very high (65%+) pre-market upvotes indicate euphoria or extreme greed. This may signal that the market/HSI is overheated and may be due for a correction soon. Likewise, extreme fear indicates an overreaction (e.g., huge selloffs), hence a subsequent undervaluation of the HSI. Buy-in signals at <35% aims to capture reversal opportunities. However, to reduce chances of a premature exit in bull market regimes, the sell-out signal is suppressed when the HSI's closing price of that day is higher than its MA30 value (indicative of a bull market).
Gap & Sentiment Gap Up/Down Strategy and MA10 of Up Vote (%)	**Buy when:** today's HSI Opening Price is higher than yesterday's Highest Price for *two consecutive* days. **Sell either when:** (1) today's HSI Opening Price is lower than yesterday's Lowest Price (2) the MA10 of Up Vote (%) reaches 60% or higher (prolonged periods of, or excessive euphoria or extreme greed).	This strategy borrows from the principles the Gap Up/Gap Down strategy. When today's Opening Price is higher than yesterday's Highest Price, it is perceived as a buy-in signal. Likewise, if today's Opening Price is lower than yesterday's Lowest Price, it is perceived as highly unusual and alerting, hence a sell-off signal. A special customization used in this report is to only issue the buy-in signal when there is two consecutive days of a Gap Up occurring. This is to ensure that there is an actual bullish momentum rather than an isolated occasion/anomaly. For more robust risk management, a sell signal is issued when there is one instead of two consecutive days of Gap Down occurs. This strategy also incorporates contrarian principles using rolling MA10 sentiment filter. When the MA10 of Up Votes reaches 60%+, a selloff signal is issued, as it indicates prolonged periods of (10 days), or excessive euphoria or extreme greed. This is to avoid losses in the case of an upcoming slump or correction due to an overheated market.

Figure 2: Definition, Logic, and Rationales of Proposed Trading Strategies

Section 3: Methodology

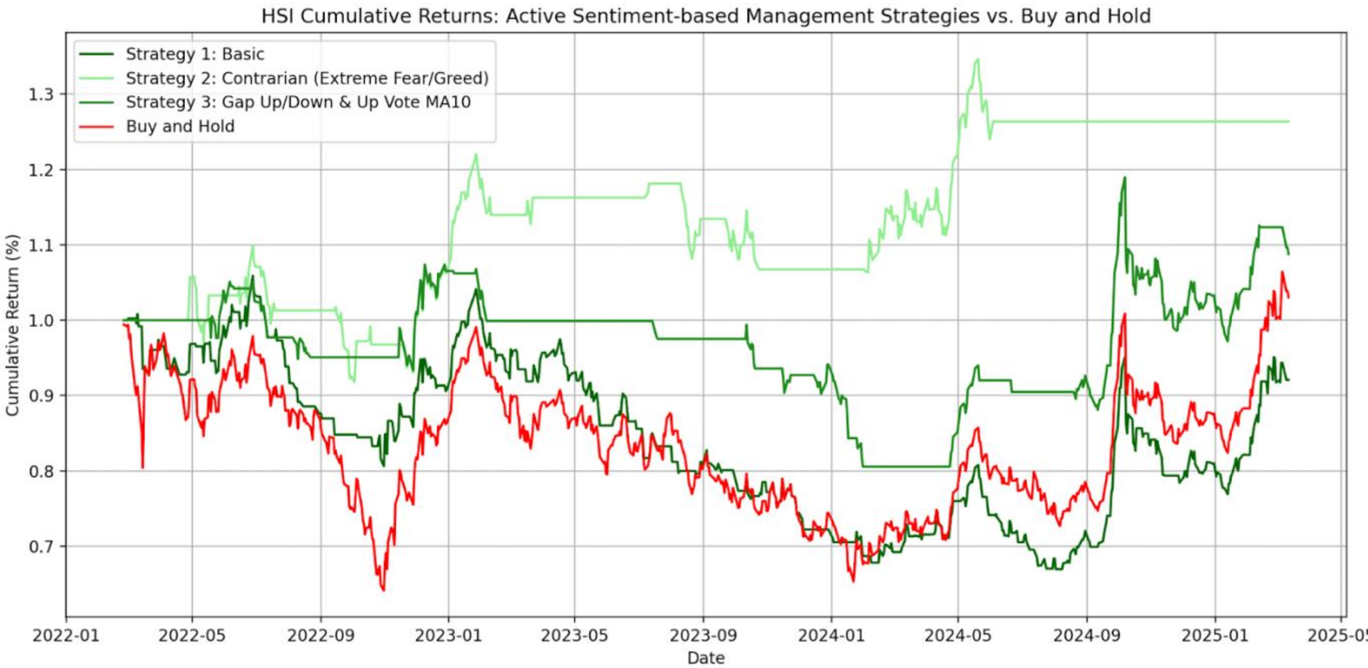
Figure 3 below outlines the key steps, frameworks and tools used in this strategy development report. Some steps are also mentioned and repeated in other sections of this report for further emphasis and remind.

Step	Description
Data Sourcing	The dataset used for this report's analysis is provided by the firm. It consists of daily social media sentiment polls collected in pre-market hours, and HSI Open, High, Low and Close prices between 24 February 2022 and 12 March 2025. Each day recorded the percentages of social media users with Up Votes and Down Votes.
Missing Data Handling	195 of the 748 rows of sentiment values were missing, accounting to 26% of the total rows. When handling missing data, the aim is to preserve data continuity while avoiding biased assumptions. A complete case analysis or mean imputation would introduce bias. Hence, the missing values were addressed using multiple imputation instead of a complete case analysis. The Multivariate Imputation by Chained Equations (MICE) method, via IterativeImputer was used to conduct multiple imputation. This approach predicts missing values based on existing values in other variables, including price-based features (Open, High, Low, Close)
Backtesting Execution	Each strategy's performance is evaluated through a backtest that cover the whole dataset period (February 2022 to March 2025). In each trading day, strategy-specific buy/sell signals are issued when sentiment and/or price action data meets the strategy's logic. For this report, a simple binary position model was used, where the strategy is either fully invested or not invested. Signals are generated with pre-market sentiment and last-day prices. Trades are assumed to execute at the opening price of the same trading day. Doing so, this ensures the backtest follows a realistic signal timing and avoids lookahead bias.
Backtesting Metrics for Performance Evaluation	Strategy performance was evaluated by its return and risk-adjusted/risk management metrics. Metrics include cumulative return, maximum drawdown, Sharpe ratio, Sortino ratio, and win rate. These were benchmarked against a passive Buy and Hold (B&H) strategy using the HSI itself as the underlying asset.
Returns Calculation Method	Returns are calculated as a daily percentage change in the HSI's closing price, on trading days. No transaction costs or slippage were assumed.

Figure 3: Data Analysis and Backtesting Methodology

Section 4: Model Selection – Back Test Results and Strategy Performance

Having developed the three trading strategies and conducted the data processing procedures necessary for the strategy backtesting, the backtest performance and visualizations for all three strategies are presented in Figure 4 below.



Strategy	Cumul. Return (%)	Max. Drawdown (%)	Win Rate (%)	Sharpe Ratio	Sortino Ratio
1-Basic	-7.92	-36.76	45.64	-0.03	-0.05
2-Contrarian	26.35	-16.58	52.17	0.63	1.00
3-Gap Up/Down & Up Vote MA10	8.79	-24.97	46.48	0.26	0.39
Buy and Hold	3.05	-35.49	46.98	0.17	0.26

Figure 4: Strategy Backtest Metrics and Visualized Performance Comparison

Seen in Figure 4, the passive 'Buy and Hold' (B&H) strategy yields a positive yet minimal cumulative return of 3.05% across the three-year period. The aim of the following analysis is to select one of the three active management strategies with the best performance by return and risk management as the firm's new HSI trading strategy.

The three strategies' performance will be benchmarked against B&H, and to one another.

4.1. Strategy 1: Performance Evaluation

Strategy 1 underperformed relative to both B&H and Strategies 2 and 3, with a negative cumulative return of -7.92% and the worst maximum drawdown at -36.76%. Although initially outperforming B&H in the first half of the analysis period, it ultimately failed. These results suggest that the strategy's simplistic reliance on daily pre-market optimism (Up Votes > 55%) are ineffective. A Sharpe ratio of -0.03 and Sortino ratio of -0.05 indicate poor reward-to-risk efficiency. These results also suggest that using daily pre-market sentiment introduces too much noise due to highly fluctuating moods, and that positive sentiment is not indicative of next day market prices.

4.2. Strategy 2: Performance Evaluation

Strategy 2 achieved a 26.35% cumulative return, nearly three times higher than Strategy 3 and 8.6 times higher than B&H. Strategy 2's overall cumulative return therefore proves the application of contrarian theory in trading the HSI largely effective. The strategy's detection of euphoric sentiment, though slightly lagged, often leads to sell-outs in early stages of market downturns, notably in July 2022, February-March 2023, and June 2024. Buy signals are also issued in the early and mid-stages of bull market regimes, such as in December 2022 and February 2024. Another outstanding performance of Strategy 2 is its 52.17%-win rate, the only strategy in this report to achieve a 50% or higher win rate.

In terms of risk-adjusted returns, Strategy 2 achieved a Sharpe Ratio of 0.63 and a Sortino Ratio of 1.00, both the highest of all other strategies. A 0.63 Sharpe Ratio score would indicate a poor risk-adjusted return in isolation. However, it is a solid, well performing score when considering the context of the HSI's B&H cumulative return of 3.05%. When annualized, the HSI's B&H return lies significantly lower than the annualized returns of risk-free assets like the US Treasury Bonds. Conversely, when adjusted for a three-year period (February 2022 to February 2025), Strategy 2's annualized return approximates 8.78%, outperforming typical U.S. Treasury bond yields (4–5%).

Strategy 2's risk management is also robust. Strategy 2's maximum drawdown was contained at -16.58%, less than half of B&H's -35.49%, suggesting a strong downside risk management. Furthermore, Strategy 2 achieved a win rate of 52.17%, the only strategy in this study to surpass the 50% threshold.

Despite its relatively strong overall performance, Strategy 2 also contains limitations. It is susceptible to occasional issuances unprofitable buy-in signals. For instance, fear-based buy-in signals were issued in market downturns in October 2022 and August 2023, leading to subsequent losses. The strategy also failed to issue buy-in signals during the dramatic bull markets in September 2024 and January 2025.

Both limitations underscore the excessive noise present in daily sentiment data and suggest that positive sentiment does not necessarily predict next-day market movements.

4.3. Strategy 3: Performance Evaluation

Strategy 3 achieved a moderate cumulative return of 8.79%, overperforming the B&H benchmark, second to Strategy 2. It is adept in capturing uptrends, which eventually lead to high returns if the uptrend develops into a bull market regime, most evident in uptrends experienced in December 2022, April 2024, and September 2024.

Strategy 3's maximum drawdown is -24.97%, again second to Strategy 2, and significantly lower than B&H's -35.49%. This suggests that Strategy 3 retains good downside risk management. Hence, whilst not as effective as Strategy 2, Strategy 3's downside risk is also lower than the benchmark B&H approach, where losses are less pronounced than B&H and Strategy 1.

An interesting observation is the premature sell-out during the December 2022 uptrend due to a high Up Vote MA10. In Strategy 3's logic, a high average sentiment (Up Vote MA10 > 60%) assumes prolonged periods of euphoria, and thereby the risks of an overheated market. This premature exit is a nuanced outcome. While bringing lower downside risk, it may also overstate euphoria risk, missing opportunities in bull markets with healthy, sustained optimism.

Other limitations of Strategy 3 include the issuance of buy-in signals during bear market regimes. Buy-in signals are when there are two consecutive gap ups, despite being in a downtrend/bear market. Therefore, the key flaw here are consecutive gap-ups can be falsely interpreted as bullish momentum. This is observed in the post-buy in losses in market downtrends in October 2022, February 2023, and October 2023 to February 2024. Finally, another limitation of Strategy 3 is its inability to exit in bear markets or bull markets with gentle down or uptrends without Gap Ups/Downs.

To maximize Strategy 3's cumulative returns, a key refinement is to define bull, bear, and sideways markets, and tighten the buy in conditions during bear market regimes.

4.4. Final recommendation

Based on the performance evaluation that assesses each strategy's cumulative return, risk-adjusted returns and downside risk management, Strategy 2 overperforms all other strategies. Hence, this report recommends Strategy 2 as the firm's new HSI trading strategy.

Section 5: Conclusion and Further Recommendations

Based on this report's backtesting results, Strategy 2, the Contrarian approach based on extreme greed or fear, generated the highest cumulative return (26.35%), strongest risk-adjusted return (Sharpe Ratio 0.63) alongside the most robust downside risk management (Maximum drawdown -16.58%). Nonetheless, its sole reliance on euphoria-based sell signals missed opportunities when positive sentiment coincided with genuine bullish uptrends. It is recommended in future refinements, to combine sentiment triggers with momentum-confirming indicators like the MACD, RSI, or moving averages.

Strategy 3, using gap up/down and euphoria detection, proved useful in bullish conditions. Yet, it is vulnerable to false positives, mistaking isolated gap-ups during bear markets as bullish signals. Enhancing regime detection to better identify broader market conditions could reduce these false positives.

Finally, Strategy 1, a basic sentiment threshold-based model, performed the worst on both return and risk dimensions and is therefore not recommended.

These findings indicate that social sentiment does indeed have predictive power and potential to develop profitable trading strategies. Whilst it also contains downsides, using it in combination with technical indicators and market regime detection may further improve its accuracy and hence, profitable strategies.

Based on the existing limitations found in this report's strategies, future strategy development should implement the following:

- Introducing moving averages or smoothing filters to reduce signal noise and better capture prevailing sentiment trends.
- Introducing bull/bear market detection filters, such as bearish MACD crosses or moving average alignment (e.g., MA2 < MA5 < MA10), to suppress false positives or risky buy signals during downtrends.
- Introducing predictive modeling to forecast future sentiment based on historical prices and the corresponding sentiment at the same day.

Appendix – Python Code

```
# HSI Social Media Sentiment-led Strategy Backtesting Report

#### IMPORTING LIBRARIES AND DATASET ####

import pandas as pd # type: ignore

# First use Pandas to load in HSI.xlsx
hsi_full_dataset = pd.read_excel('/Users/lambchops7379/Desktop/DataLouder Backtesting/HSI.xlsx')

# Find no. of missing values
print('')
print('*** Missing Values Overview ***')
print('')

print(hsi_full_dataset.isnull().sum()) # There are 195 missing sentiment/votes values

print('')
print('*** HSI.xlsx Dataset Overview ***')
print('')

print(hsi_full_dataset)


#### MULTIPLE IMPUTATION FOR MISSING SENTIMENT DATA ####

# Multiple Imputation (MI) variables
data_for_mi = hsi_full_dataset[['Open', 'High', 'Low', 'Close', 'Up votes', 'Down votes']].copy()

from sklearn.experimental import enable_iterative_imputer # type: ignore
from sklearn.impute import IterativeImputer

# Initialize the MICE-based imputer
imp = IterativeImputer(max_iter=10, random_state=0)

# Fit and transform
imputed_array = imp.fit_transform(data_for_mi)
```

```
# Convert back to DataFrame
imputed_dataset = pd.DataFrame(imputed_array, columns=data_for_mi.columns, index=data_for_mi.index)

# Update original dataset
hsi_full_dataset['Up votes'] = imputed_dataset['Up votes'].round(2)
hsi_full_dataset['Down votes'] = imputed_dataset['Down votes'].round(2)


hsi_full_dataset.to_csv("/Users/lambchops7379/Desktop/HSI_Imputed_new.csv", index=True)



#### INITIAL DATA AND TREND VISUALIZATION ####

# Initial data and trend exploration

# Import the imputed dataset
import matplotlib.pyplot as plt
from matplotlib.patches import Rectangle
import matplotlib.dates as mdates
import pandas as pd
import datetime

hsi_imputed_dataset = pd.read_csv('/Users/lambchops7379/Desktop/DataLouder Backtesting/HSI_Imputed_new.csv')

print('')
print('*** HSI Imputed Dataset Overview ***')
print('')

print(hsi_imputed_dataset)

# Convert 'Date' into date format
hsi_imputed_dataset['Date'] = pd.to_datetime(hsi_imputed_dataset['Date'])
hsi_imputed_dataset.set_index('Date', inplace = True)


hsi_imputed_dataset['Total votes'] = hsi_imputed_dataset['Up votes'] + hsi_imputed_dataset['Down votes']
```

```
hsi_imputed_dataset['Up_percent'] = hsi_imputed_dataset['Up votes'] / hsi_imputed_dataset['Total votes'] * 100
hsi_imputed_dataset['Down_percent'] = 100 - hsi_imputed_dataset['Up_percent']


# Moving Average (MA) 3, 5, 10, 30 for Up Vote (%)
hsi_imputed_dataset['Up_percent_MA2'] = hsi_imputed_dataset['Up_percent'].rolling(2).mean()
hsi_imputed_dataset['Up_percent_MA5'] = hsi_imputed_dataset['Up_percent'].rolling(5).mean()
hsi_imputed_dataset['Up_percent_MA10'] = hsi_imputed_dataset['Up_percent'].rolling(10).mean()
hsi_imputed_dataset['Up_percent_MA30'] = hsi_imputed_dataset['Up_percent'].rolling(30).mean()

fig, (ax1, ax2) = plt.subplots(
    2, 1, figsize=(20, 8), sharex=True,
    gridspec_kw={'height_ratios': [1, 1]}
)
ax1.plot(hsi_imputed_dataset.index, hsi_imputed_dataset['Up_percent'], alpha=0.3, label='Up vote %', color='gray')
ax1.plot(hsi_imputed_dataset.index, hsi_imputed_dataset['Up_percent_MA2'], label='2-Day MA', color='blue')
ax1.plot(hsi_imputed_dataset.index, hsi_imputed_dataset['Up_percent_MA5'], label='5-Day MA', color='green')
ax1.plot(hsi_imputed_dataset.index, hsi_imputed_dataset['Up_percent_MA10'], label='10-Day MA', color='orange')

ax1.set_ylabel('Up Vote %')
ax1.set_title('Daily Up Vote % with 2, 5 & 10-Day Moving Averages')
ax1.legend()
ax1.grid(True)

# Plot 2: Plotting HSI data
dates = mdates.date2num(hsi_imputed_dataset.index.to_pydatetime())

for i in range(len(hsi_imputed_dataset)):
    # Coding red and green candles
    open, high, low, close = hsi_imputed_dataset.iloc[i][['Open', 'High', 'Low', 'Close']]
    color = 'green' if close >= open else 'red'
    ax2.add_patch(Rectangle((dates[i] - 0.05, min (open, close)), 0.3, abs(open - close), color = color))
    ax2.plot([dates[i], dates[i]], [low, high], color = color)


import datetime
```

```
# Set X-axis limits closer to dataset's actual start/end to reduce white space
ax2.set_xlim(datetime.datetime(2022, 2, 14), datetime.datetime(2025, 3, 22))

# X-axis as date
ax2.xaxis_date()

ax2.xaxis.set_major_formatter(mdates.DateFormatter('%Y-%m'))
plt.xticks(rotation=45)
plt.title('HSI Daily Candlestick Chart: 24 February 2022 to 12 March 2025')

# X & Y axis label names
plt.xlabel("Date")
plt.ylabel("Price")

# Add grids to graph
plt.grid(True)
plt.tight_layout()

plt.show()


#### STRATEGY DESIGN/DEVELOPMENT ####


### STRATEGY 1: Basic model - Majority Up Vote = Buy, Less than Majority = Sell ###

# Calculate daily returns

hsi_imputed_dataset['Return'] = hsi_imputed_dataset['Close'].pct_change()

# Strategy 1: Basic
hsi_imputed_dataset['Strat1_Position'] = 0
holding = False

for i in range(1, len(hsi_imputed_dataset)):
    upvote = hsi_imputed_dataset['Up votes'].iloc[i]
```

```
    if upvote > 0.55 and not holding:
        hsi_imputed_dataset.at[hsi_imputed_dataset.index[i], 'Strat1_Position'] = 1
        holding = True
    elif upvote < 0.50 and holding:
        hsi_imputed_dataset.at[hsi_imputed_dataset.index[i], 'Strat1_Position'] = 0
        holding = False
    else:
        hsi_imputed_dataset.at[hsi_imputed_dataset.index[i], 'Strat1_Position'] = hsi_imputed_dataset['Strat1_Position'].iloc[i - 1]


hsi_imputed_dataset['Strategy1_Return'] = hsi_imputed_dataset['Strat1_Position'].shift(1) * hsi_imputed_dataset['Return']

# Strategy 1 cumulative returns
hsi_imputed_dataset['Cumulative_Strategy1'] = (1 + hsi_imputed_dataset['Strategy1_Return']).cumprod()
hsi_imputed_dataset['Cumulative_BuynHold'] = (1 + hsi_imputed_dataset['Return']).cumprod()


# ### STRATEGY 2 ENHANCED: Extreme Fear and Greed, higher selling thereshold in Bull Market ###

print('')
print('Strategy 2: Contrarian with higher bull market threshold')
print('')

hsi_imputed_dataset['Return'] = hsi_imputed_dataset['Close'].pct_change()

# Strategy 2 Enhanced: Extreme Fear / Greed
hsi_imputed_dataset['Strat2e_Position'] = 0
holding = False

# Add bull market threshold (30-day moving average)
hsi_imputed_dataset['MA30'] = hsi_imputed_dataset['Close'].rolling(window=30).mean()
ma30 = hsi_imputed_dataset['MA30']

# Strategy 2: Extreme Fear Or Greed with Bull Market Threshold

for i in range(30, len(hsi_imputed_dataset)):
    closing_price = hsi_imputed_dataset['Close'].iloc[i]  # current close price
```

```
    upvote = hsi_imputed_dataset['Up votes'].iloc[i]

    if upvote <= 0.35 and not holding:
        hsi_imputed_dataset.at[hsi_imputed_dataset.index[i], 'Strat2e_Position'] = 1
        holding = True
    elif upvote >= 0.65 and closing_price < ma30.iloc[i] and holding:
        hsi_imputed_dataset.at[hsi_imputed_dataset.index[i], 'Strat2e_Position'] = 0
        holding = False
    else:
        hsi_imputed_dataset.at[hsi_imputed_dataset.index[i], 'Strat2e_Position'] = hsi_imputed_dataset['Strat2e_Position'].iloc[i -
1]


# Shift strategy position by 1 to avoid lookahead bias
hsi_imputed_dataset['Strategy2e_Return'] = hsi_imputed_dataset['Strat2e_Position'].shift(1) * hsi_imputed_dataset['Return']

# Strategy 2 Enhanced: cumulative returns vs Buy and Hold
hsi_imputed_dataset['Cumulative_Strategy2e'] = (1 + hsi_imputed_dataset['Strategy2e_Return']).cumprod()
hsi_imputed_dataset['Cumulative_BuynHold'] = (1 + hsi_imputed_dataset['Return']).cumprod()


# STRATEGY 3: Gap Up/Down and MA10 Sentiment
# Ensure date index is sorted
hsi_imputed_dataset = hsi_imputed_dataset.sort_index()

# Compute previous high and low
hsi_imputed_dataset['Prev_High'] = hsi_imputed_dataset['High'].shift(1)
hsi_imputed_dataset['Prev_Low'] = hsi_imputed_dataset['Low'].shift(1)

# Compute MA10 of Upvote %
hsi_imputed_dataset['Upvote_MA10'] = hsi_imputed_dataset['Up_percent'].rolling(window=10).mean()

# Initialize columns
hsi_imputed_dataset['Buy_Signal_3'] = False
hsi_imputed_dataset['Sell_Signal_3'] = False
hsi_imputed_dataset['Strategy3_Position'] = 0
```

```python
# Strategy logic
holding = False
buy_streak = 0
sell_streak = 0

for i in range(1, len(hsi_imputed_dataset)):
    today = hsi_imputed_dataset.index[i]
    yesterday = hsi_imputed_dataset.index[i - 1]

    open_today = hsi_imputed_dataset.at[today, 'Open']
    high_yesterday = hsi_imputed_dataset.at[yesterday, 'High']
    low_yesterday = hsi_imputed_dataset.at[yesterday, 'Low']
    ma10 = hsi_imputed_dataset.at[today, 'Upvote_MA10']

    # Buy streak
    if open_today > high_yesterday:
        buy_streak += 1
    else:
        buy_streak = 0

    # Sell streak
    if open_today < low_yesterday:
        sell_streak += 1
    else:
        sell_streak = 0

    # Conditions for Buy in
    if not holding and buy_streak >= 2:
        hsi_imputed_dataset.at[today, 'Buy_Signal_3'] = True
        hsi_imputed_dataset.at[today, 'Strategy3_Position'] = 1
        holding = True
        sell_streak = 0
        continue

    # Conditions for Selling
    if holding and (sell_streak >= 2 or ma10 > 60):
        hsi_imputed_dataset.at[today, 'Sell_Signal_3'] = True
```

```
        hsi_imputed_dataset.at[today, 'Strategy3_Position'] = 0
        holding = False
        buy_streak = 0
        sell_streak = 0
    else:
        hsi_imputed_dataset.at[today, 'Strategy3_Position'] = hsi_imputed_dataset.at[yesterday, 'Strategy3_Position']

hsi_imputed_dataset['Strategy3_Return'] = hsi_imputed_dataset['Strategy3_Position'].shift(1) * hsi_imputed_dataset['Return']
hsi_imputed_dataset['Cumulative_Strategy3'] = (1 + hsi_imputed_dataset['Strategy3_Return']).cumprod()



# Calculate strategy returns
hsi_imputed_dataset['Strategy3_Return'] = hsi_imputed_dataset['Strategy3_Position'].shift(1) * hsi_imputed_dataset['Return']
hsi_imputed_dataset['Cumulative_Strategy3'] = (1 + hsi_imputed_dataset['Strategy3_Return']).cumprod()


#### COMBINED GRAPH ####
plt.figure(figsize=(12, 6))

# Plot each strategy's cumulative return
plt.plot(hsi_imputed_dataset['Cumulative_Strategy1'], label='Strategy 1: Basic', color='#006400')
plt.plot(hsi_imputed_dataset['Cumulative_Strategy2e'], label='Strategy 2: Contrarian (Extreme Fear/Greed)', color='#90EE90')
plt.plot(hsi_imputed_dataset['Cumulative_Strategy3'], label='Strategy 3: Gap Up/Down & Up Vote MA10',color='#228B22')
plt.plot(hsi_imputed_dataset['Cumulative_BuynHold'], label='Buy and Hold', color = 'red')

plt.legend()
plt.title('HSI Cumulative Returns: Active Sentiment-based Management Strategies vs. Buy and Hold')
plt.xlabel('Date')
plt.ylabel('Cumulative Return (%)')
plt.grid(True)
plt.tight_layout()
plt.show()




####### BACKTESTING METRICS #######
```

```python
import quantstats as qs  # type: ignore
qs.extend_pandas()


# Strategy 1: Basic
cumulative_return_1 = hsi_imputed_dataset['Cumulative_Strategy1'].iloc[-1] - 1
sharpe_ratio_1 = hsi_imputed_dataset['Strategy1_Return'].mean() / hsi_imputed_dataset['Strategy1_Return'].std() * (252 ** 0.5)
max_drawdown_1 = (hsi_imputed_dataset['Cumulative_Strategy1'] / hsi_imputed_dataset['Cumulative_Strategy1'].cummax() -
1).min()
sortino_ratio_1 = hsi_imputed_dataset['Strategy1_Return'].dropna().sortino()


print('')
print('*** Backtesting measures for Strategy 1: Basic ***')
print(f"Cumulative Return: {cumulative_return_1:.2%}")
print(f"Sharpe Ratio: {sharpe_ratio_1:.2f}")
print(f"Max Drawdown: {max_drawdown_1:.2%}")
print(f"Sortino Ratio: {sortino_ratio_1:.2f}")
print(f"Win Rate: {hsi_imputed_dataset['Strategy2e_Return'].dropna().win_rate():.2%}")




# Strategy 2: Extreme Fear/Greed
cumulative_return_2 = hsi_imputed_dataset['Cumulative_Strategy2e'].iloc[-1] - 1
sharpe_ratio_2 = hsi_imputed_dataset['Strategy2e_Return'].mean() / hsi_imputed_dataset['Strategy2e_Return'].std() * (252 **
0.5)
max_drawdown_2 = (hsi_imputed_dataset['Cumulative_Strategy2e'] / hsi_imputed_dataset['Cumulative_Strategy2e'].cummax()
- 1).min()
sortino_ratio_2 = hsi_imputed_dataset['Strategy2e_Return'].dropna().sortino()

print('')
print('*** Backtesting measures for Strategy 2: Ext reme Fear/Greed ***')
print('')

print(f"Cumulative Return: {cumulative_return_2:.2%}")
print(f"Sharpe Ratio: {sharpe_ratio_2:.2f}")
print(f"Max Drawdown: {max_drawdown_2:.2%}")
```

```python
print(f"Sortino Ratio: {sortino_ratio_2:.2f}")
print(f"Win Rate: {hsi_imputed_dataset['Strategy2e_Return'].dropna().win_rate():.2%}")


# Strategy 3: Gap Up/Down & Up Vote MA10
cumulative_return_3 = hsi_imputed_dataset['Cumulative_Strategy3'].iloc[-1] - 1
sharpe_ratio_3 = hsi_imputed_dataset['Strategy3_Return'].mean() / hsi_imputed_dataset['Strategy3_Return'].std() * (252 ** 0.5)
max_drawdown_3 = (hsi_imputed_dataset['Cumulative_Strategy3'] / hsi_imputed_dataset['Cumulative_Strategy3'].cummax() -
1).min()
sortino_ratio_3 = hsi_imputed_dataset['Strategy3_Return'].dropna().sortino()

print('')
print('*** Strategy 3: Gap Up/Down & Up Vote MA10 ***')
print('')

print(f"Cumulative Return: {cumulative_return_3:.2%}")
print(f"Sharpe Ratio: {sharpe_ratio_3:.2f}")
print(f"Max Drawdown: {max_drawdown_3:.2%}")
print(f"Sortino Ratio: {sortino_ratio_3:.2f}")
print(f"Win Rate: {hsi_imputed_dataset['Strategy3_Return'].dropna().win_rate():.2%}")



# Passive: Buy and Hold
cumulative_return_bh = hsi_imputed_dataset['Cumulative_BuynHold'].iloc[-1] - 1
sharpe_ratio_bh = hsi_imputed_dataset['Return'].mean() / hsi_imputed_dataset['Return'].std() * (252 ** 0.5)
max_drawdown_bh = (hsi_imputed_dataset['Cumulative_BuynHold'] / hsi_imputed_dataset['Cumulative_BuynHold'].cummax() -
1).min()
sortino_bh = hsi_imputed_dataset['Return'].dropna().sortino()

print('')
print('*** Backtesting measures for Buy and Hold ***')
print('')

print(f"Cumulative Return: {cumulative_return_bh:.2%}")
print(f"Sharpe Ratio: {sharpe_ratio_bh:.2f}")
print(f"Max Drawdown: {max_drawdown_bh:.2%}")
```

```
print(f"Win Rate: {hsi_imputed_dataset['Cumulative_BuynHold'].dropna().win_rate():.2%}")
print(f"Sortino Ratio: {sortino_bh:.2f}")
```